

Customizable Multi-Modal Mixed Reality Framework

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ABSTRACT

Mixed Reality (MR) has potential to be used not only in the entertainment industry and the work industry, but also for assistive technology. Mixed Reality is suited for assistive technology because of its utilization of other forms of physical feedback to the human senses besides vision while still making use of visual feedback. We propose a glove-based MR system framework that will use finger and hand movement tracking along with tactile feedback so that the blind and visually-impaired (BVI) can interact tactily with virtual objects. In addition to touch, our proposed framework will include robust interactions through other modalities such as a custom voice assistant and an audio interface as well as visual interfaces that tailor to the visual needs of BVI users. Through the various modalities of interaction in our proposed framework, BVI users will be able to obtain a more detailed sense of virtual objects of any 3D model and their experiences will not be limited by vision. The customizable features and modalities that will be available in our proposed system framework will allow for a more individual experience that can be tailored to the variety of the different needs of the BVI as well as general users.

Index Terms: Extended Reality—Virtual Reality—Mixed Reality—Assistive Technology—Blind and Visually Impaired—Accessibility—User Interface—Haptic Feedback

1 INTRODUCTION

The overarching domain of Extended Reality (XR) and its subsets of Virtual Reality (VR), Augmented Reality (AR), and Mixed Reality (MR) are research areas of growing interest due to the extensive amount of potential applications across many industries such as education, work and entertainment. One particular type of application of XR is assistive technology. There are many works in academia that explored the design and usage of MR systems in particular for the blind and visually impaired (BVI) [2, 4, 5]. However, the works are primarily for navigation guidance for the visually impaired. Though this is an important task and application area for BVI, another important application of MR for BVI is object exploration and interaction. Creating an enriching and accommodating experience for this application is the goal of our research.

Our proposed MR system framework will include implementations for tactile interaction through haptic feedback, an audio user interface (UI) and a visual user interface. For tactile interaction, the hardware for our MR system framework will be embedded in a glove-based system. The tactile modality will include finger and hand movement tracking to move the virtual hand accordingly and vibromotors to provide haptic feedback once their corresponding virtual fingertips touch a virtual object. In our proposed audio user interface, there is a STT module for receiving commands and a TTS module for relaying verbal feedback to the user. The GUI will include augmentations to the virtual objects for interaction and exploration as well as various user controls and information.

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The visual interfacing of XR systems almost seems to be an inherent part of the domain technology, and although it is inarguably a valuable aspect of the system, too much exclusive reliance on the modality can be a limitation to BVI users. Many times the visual interfacing doesn't consider or accommodate for any of the visual needs of BVI users. There is a common misconception that most BVI can't see at all, but in reality BVI that are completely blind are actually rare and most BVI have a significant amount of residual vision that they even prefer to rely on over other senses when possible [1]. This means that contrary to the common assumption, graphical user interfaces (GUIs) and other visual aspects of XR are still important considerations for assistive MR system design. This means that when we design our XR application with accessibility in mind, we are investigating different accommodating visual features such as clear and contrasting colors and larger font sizes and further customizability in the user interface. We are also researching the usage of computer vision to extract information about objects that could normally be obtained quickly through visual inspection. Beyond just object classification, we are researching what information will be useful to extract and then provide to BVI users.

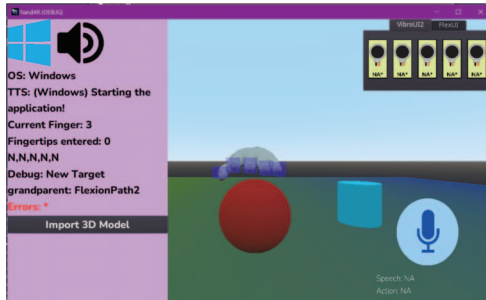
Our proposed system also will make use of haptic feedback to emulate the feeling of physical touch and essentially allow the user to feel and interact with virtual objects that aren't actually physically there. Tactile interaction can help BVI get a better sense of the shapes and other visual features of different objects that can be represented virtually and thus interacted with through MR systems. The tactile modality is also beneficial for interacting with objects that would be unsafe in the actual physical space or otherwise inaccessible even to general users. Though tactile interaction is a valuable modality, it can be further utilized by combining other modalities of interaction to enhance the overall experience and give it more depth.

Another modality that can be utilized to overcome limitations in vision is audio. Though the BVI still can make use of GUIs when they can meet enough of their visual needs, some BVI still want or even prefer to use audio interfaces such as screenreaders or voice assistants. Two important features of the audio modality are Speech to Text (STT) and Text to Speech (TTS). Our system framework utilizes STT to receive commands from the user and translate them to actions in the system. We are also researching natural language processing (NLP) for usage in conjunction with STT for more flexible and natural commands rather than limiting the interaction to strict keywords for the user to be restricted to. In addition, TTS is also used in our system framework in conjunction with the GUI to give audio feedback/an audio counterpart for every corresponding visual GUI element and action. For example, when the user first touches the virtual object, an audio message with confirmation of touching as well as some of the properties of the virtual object will be verbally relayed to the user. The TTS will also be used to give verbal feedback to the user's verbal STT commands to create a primitive form of a voice with dialog. Using audio and visual modalities in conjunction can allow the user to pick whatever feedback combination suits them best rather than only have one or the other.

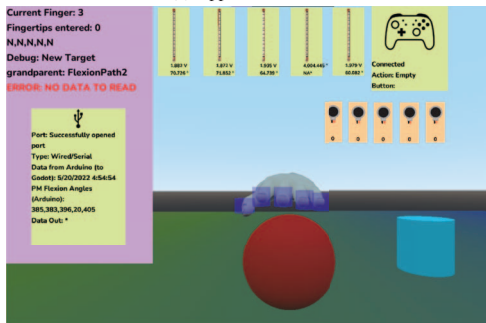
There is a vast online database of 3D models of various things such as furniture, commodities, and even things that don't actually exist and are fictional like dragons or other fictional objects. There are 3D models for things that are normally not accessible without

vision or are very limited in accessibility to through other modalities, such as models for large structures like entire buildings or landmarks or the opposite, small structures like atoms and molecules. With our proposed system framework, this vast database can be utilized to allow robust multi-modal interaction with a 3D model of any object of a user's choosing.

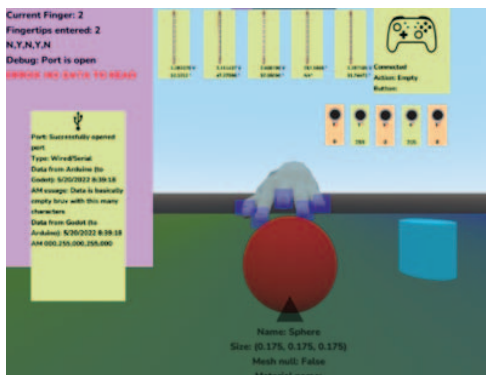
2 PRELIMINARY RESULTS



(a) Application GUI.



(b) The virtual environment along with the sensor GUI with numeric indicators for the corresponding finger's sensor data being acquired in real time.



(c) The virtual environment along with the vibromotor menu, toggled on, with visual and numeric indicators for the corresponding finger's vibromotor being turned on.

Figure 1: System Prototype.

In our glove system prototype, which we have implemented in our previous work [3], we use flex sensors for the finger flexion and extension movement tracking and are planning to also incorporate inertial measurement units (IMUs) for the tracking of the other types of finger and hand movements. The prototype also has a vibromotor and representative red LEDs for visualization purposes as vibrations cannot be seen in camera images. For testing purposes, we also have other forms of inputs such as a videogame controller and keyboard to control the movement of the virtual hand. In Fig. 1b, the flex sensor

is being flexed in the circuit and, in real time, the sensor GUI shows the flex sensor resistance data as well as the resistance converted to the corresponding flexion angle.

In our virtual environment application, which can be hosted on both Windows and Android OS, we first have the virtual hand with the fingertips encased in purple volumes for visualization purposes. Then we have virtual objects of a sphere and a cylinder. Toggled on are the sensor and vibromotor GUIs in which we can see live data being shown and indicative background colors. We also have preliminary implementations of a voice assistant UI as well as a UI for importing any 3D model into the virtual environment for exploration and are working to develop them more further. Although the exploration environment is currently entirely in a virtual setting, we are also developing to have a mode to augment the virtual objects into a visual augmentation of a camera view.

As shown in Fig. 1c, when the prototype system's inputs move the virtual hand so that virtual object is touched, the vibromotor interfaces of the fingers that are touching the virtual object show both the pulse width modulation (PWM) value as well as a green background rather than the default background. The corresponding output LEDs are also turned on as well as the vibromotor vibrating, verifying the haptic feedback of the interaction. Some of the virtual object properties that we have set such as the virtual object's name and size are also shown visually through text and spoken through the TTS. The object properties that are relayed to the user can be modified and expanded based upon studies we could conduct into what information would be useful to relay to BVI users.

3 CONCLUSION

With our research, we are targeting a philanthropic application and want to give BVI a more enriched and comprehensive experience that they can tailor to their different needs. The inclusively multi-modal nature of our proposed framework can also have beneficial usages in other domains and applications and enhance the experiences of those applications as well. One of the areas we are looking for feedback is the efficiency of the hardware in the glove system and the benefits that efficiency can give for the target application. Another area of potential improvement is overall application design and how we can adhere to existing standards for more seamless integration with established technologies and easy usability.

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